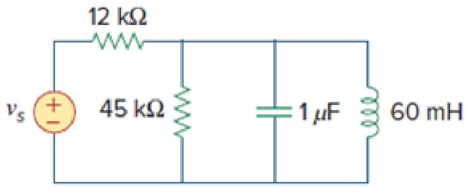
Chapter 14, Problem 29P

Problem

Let *vs* = 20 cos(*at*) V in the circuit of Fig. 14.77. Find *ω*0, *Q*, and *B*, as seen by the capacitor.

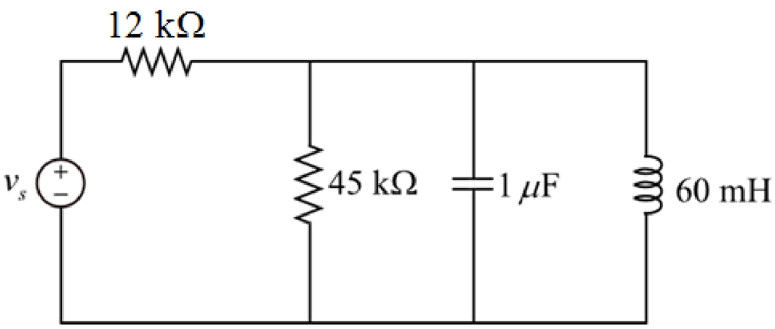
Figure 14.77:



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram:



Step 2 of 7

Figure 1

Step 3 of 7

Consider the following data:

Input voltage $v_s = 20\cos(at)$ V.

Step 4 of 7

A source transformation is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R, or vice versa.

Apply source transformation to the circuit, we will get current source i_s in parallel with a resistor $12\text{ k}\Omega$ as shown below.

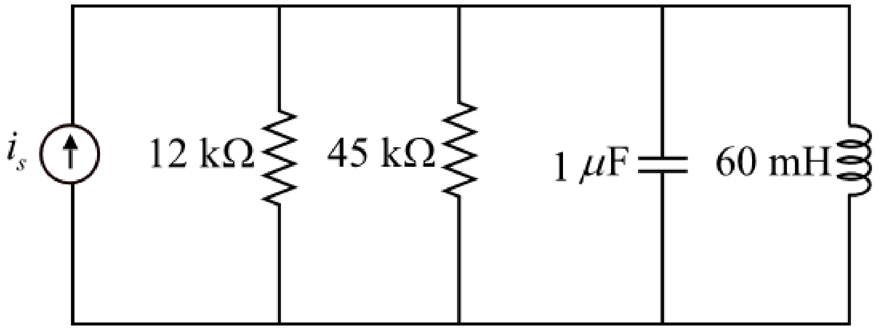


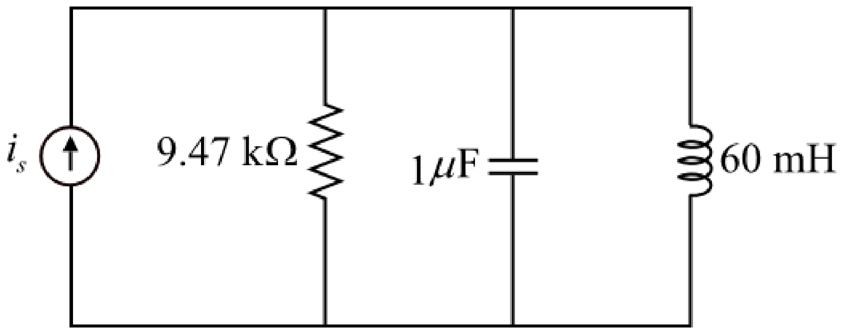
Figure 2

Step 5 of 7

To get equivalent resistance R, we combine resistors in parallel and in series. The $45\text{ k}\Omega$ and $12\text{ k}\Omega$ resistors are in parallel. So their equivalent resistnace is:

$$\begin{aligned} R &= 45\text{ k}\Omega \| 12\text{ k}\Omega \\ &= \frac{45\text{ k} \times 12\text{ k}}{45\text{ k} + 12\text{ k}} \\ &= 9.47\text{ k}\Omega \end{aligned}$$

Redraw, figure 2 by replacing resistors $45\text{ k}\Omega$ and $12\text{ k}\Omega$ with equivalent resistance $9.47\text{ k}\Omega$.



Step 6 of 7

Figure 3 is a parallel RLC circuit. Thus

Write the expression for resonant frequency ω_0.

$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(60\times10^{-3})(1\times10^{-6})}} \\ &= \frac{1}{\sqrt{6\times10^{-8}}} \\ &= \frac{1}{0.244\times10^{-3}} \\ &= 4082.48\text{ rad/s} \\ &= 4.082\text{ krad/s} \end{aligned}$$

Thus, resonant frequency ω_0 is $\boxed{4.082\text{ krad/s}}$.

Write the expression for quality factor of a parallel resonance circuit:

$$Q = \omega_0 RC$$

Substitute $9.47\ \text{k}\Omega$ for R, $1\ \mu$ for C and 4082.48 for ω_0 in the expression $\omega_0 RC$ to get the quality factor.

$$\begin{aligned} Q &= (4082.4)\left(9.47\times10^{3}\right)\left(1\times10^{-6}\right) \\ &\cong 38.67 \end{aligned}$$

Therefore the quality factor Q is $\boxed{38.67}$.

Write the expression for bandwidth of the circuit.

$$B = \frac{\omega_0}{Q}$$

Substitute 38.67 for Q and 4082.48 for ω_0 values in the above expression.

$$\begin{aligned} B &= \frac{4082}{38.67} \\ &= 105.55\ \text{rad/s} \end{aligned}$$

Therefore the bandwidth of the circuit is $\boxed{105.55\ \text{rad/s}}$.